

Semester - III			
Sl. No.	Course Code	Course Title	Page No.
1	MA231TD	Applied Mathematics for Civil Engineering	1
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	3 / 5 / 7
3	CV234AI	Concrete Technology	9
4	CV235AI	Mechanics of Materials	11
5	CV236AI	Building-Systems, Construction, Planning	13
6	CV237AT	AI for Civil Engineering	15
7	CV238DL	Applied Design thinking lab	17
8	MAT149AT	Bridge Course Mathematics	18

Semester - IV			
Sl. No.	Course Code	Course Title	Page No.
1	CS241AT	Discrete mathematical structures and combinatorics	19
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	3 / 5 / 7
3	CD343AI	Design and Analysis of Algorithms	21
4	CS345AI	Embedded Systems, IoT and Edge AI	23
5	CS246AT	Computer Networks	25
6	CS248ATX	Professional Electives	27
7	HS248LX	AE Course	39
8	CS139AT	Bridge Course C Programming	41



2025 Scheme template – UG Programs

B.E in Civil Engineering (CV)

III SEMESTER

Sl.	Course Code	Course Title	BoS	Credit Allocation				Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE	
				L	T	P	Total			Theory	Lab		Theory	Lab
1	MA231TD	Applied Mathematics for Civil Engineering	MA	3	1	0	4	Theory	2	100	***	3	100	***
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	CV / ME / BT	3	0	0	3	Theory	2	100	***	3	100	***
3	CV234AI	Concrete Technology	CV	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
4	CV235AI	Mechanics of Materials	CV	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
5	CV236AI	Building-Systems, Construction, Planning	CV	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
6	CV237AT	AI for Civil Engineering	CS	3	0	0	3	Theory	2	100	***	3	100	***
7	CV238DL	Applied Design thinking lab	CV	0	0	1	1	Lab	1.5	***	50	2	***	50
8	MAT149AT	Bridge Course Mathematics	MA	3	0	0	0*	Theory	2	100	***	—	***	***
Total				21	1	4	23							

Note: 1 Credit = 30 hrs (L:T:P)

* Audit



2025 Scheme template – UG Programs

B.E in Civil Engineering (CV)

IV SEMESTER

Sl.	Course Code	Course Title	BoS	Credit Allocation				Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE	
				L	T	P	Total			Theory	Lab		Theory	Lab
1	CS241AT	Discrete mathematical structures and combinatorics	CS	3	0	0	3	Theory	2	100	***	3	100	***
2	CV232TA / ME232TB / BT232TD	Environment and Sustainability / Material Science for Engineers / Bioinspired Engineering	CV / ME / BT	3	0	0	3	Theory	2	100	***	3	100	***
3	CD343AI	Design and Analysis of Algorithms	CD	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
4	CS345AI	Embedded Systems, IoT and Edge AI	CS	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
5	CS246AT	Computer Networks	CS	3	1	0	4	Theory	2	100	***	3	100	***
6	CS248ATX	Professional Electives	CS	3	0	0	3	Theory	2	100	***	3	100	***
7	HS248LX	AE Course	HS	0	0	1	1	Lab	1.5	***	50	2	***	50
8	CS139AT	Bridge Course C Programming	CS	3	0	0	0*	Theory	2	100	***	—	***	***
Total				21	1	3	22							

Note: 1 Credit = 30 hrs (L:T:P)

* Audit

Semester - IV			
Professional Electives			
GROUP-A			
Sl. No.	Course Code	Course Title	Page No.
1	CS248ATA	Graph Theory	27
2	CS248ATB	Computer Graphics and Extended Reality	29
3	CI248ATC	Quantum Algorithms and Key Distribution	31
4	CS248ATD	Natural Language Processing	33
5	CS355TBF	Linux Internals	35
6	CS365TDB	Operation Research	37

Semester - IV			
AE Courses			
Sl. No.	Course Code	Course Title	Page No.
1	HS238LA	National Service Scheme	39
2	HS238LB	National Cadet Corps	40



Semester - III					
Applied Mathematics for Civil Engineering Category: Professional Core Course (Theory)					
Course Code	:	MA231TD	CIE	:	100 Marks
Credit distribution L:T:P	:	3:1:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:30:0:45	CIE Duration	:	2 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - III					
Environment and Sustainability Category: Professional Core Course (Theory)					
Course Code	:	CV232TA	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - III					
Material Science for Engineers Category: Professional Core Course (Theory)					
Course Code	:	ME232TB	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - III					
Bioinspired Engineering Category: Professional Core Course (Theory)					
Course Code	:	BT232TD	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - III					
Concrete Technology Category: Professional Core Course (Theory & Practical)					
Course Code	:	CV234AI	CIE	:	50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	2+2 Hours
Total credits:	:	4	SEE Duration	:	3+3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and Problem based learning review phase I and II (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE FOR 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	50
TOTAL		50



Semester - III					
Mechanics of Materials Category: Professional Core Course (Theory & Practical)					
Course Code	:	CV235AI	CIE	:	50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	2+2 Hours
Total credits:	:	4	SEE Duration	:	3+3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and Problem based learning review phase I and II (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE FOR 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	50
TOTAL		50



Semester - III					
Building-Systems, Construction, Planning Category: Professional Core Course (Theory & Practical)					
Course Code	:	CV236AI	CIE	:	50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	2+2 Hours
Total credits:	:	4	SEE Duration	:	3+3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and Problem based learning review phase I and II (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE FOR 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	50
TOTAL		50



Semester - III					
AI for Civil Engineering Category: Professional Core Course (Theory)					
Course Code	:	CV237AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - III					
Applied Design thinking lab Category: PO (Practical)					
Course Code	:	CV238DL	CIE	:	50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	1.5 Hours
Total credits:	:	1	SEE Duration	:	2 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	Presentation 1: Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data.	10
2	EXPERIENTIAL LEARNING: Presentation 2 (phase 2) Content development, strategies for implementation methodologies (10), Case Study-based Teaching-Learning (10), Sector wise study and consolidation (10), Video based seminar (4-5 minutes per student) (10), TOTAL MARKS ADDED UPTO 40 MARKS	40
MAXIMUM MARKS FOR THE CIE		50

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Implementation strategies of the project with report	50
TOTAL		50



Semester - III					
Bridge Course Mathematics Category: Bridge Course (Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	MAT149AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	— Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	— Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100



Semester - IV					
Discrete mathematical structures and combinatorics Category: Professional Core Course (Common to Common to CS and CY Programs) (Theory)					
Course Code	:	CS241AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
Design and Analysis of Algorithms Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Theory & Practical)					
Course Code	:	CD343AI	CIE	:	50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	2+2 Hours
Total credits:	:	4	SEE Duration	:	3+3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and Problem based learning review phase I and II (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE FOR 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	50
TOTAL		50



Semester - IV					
Embedded Systems, IoT and Edge AI Category: Professional Core Course (Common to CS and CI Programs) (Theory & Practical)					
Course Code	:	CS345AI	CIE	:	50 Marks
Credit distribution L:T:P	:	3:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45	CIE Duration	:	2+2 Hours
Total credits:	:	4	SEE Duration	:	3+3 Hours
Unit - I					4 Hours
sample					
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20). ADDING UPTO 40 MARKS.	40
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and Problem based learning review phase I and II (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE FOR 50 MARKS.	50
MAXIMUM MARKS FOR THE CIE		150



RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	50
TOTAL		50



Semester - IV					
Computer Networks					
Category: Professional Core Course (Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	CS246AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:1:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:30:0:45	CIE Duration	:	2 Hours
Total credits:	:	4	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY & TUTORIAL) (Continued)		
Q.NO.	CONTENTS	MARKS
3	Unit - II: Compulsory	18
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
Graph Theory Category: Professional Core Elective (Only CS) (Theory)					
Course Code	:	CS248ATA	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
Computer Graphics and Extended Reality Category: Professional Core Elective (Only CS) (Theory)					
Course Code	:	CS248ATB	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
Quantum Algorithms and Key Distribution Category: Professional Core Elective (Common to CS, CI and CD Programs) (Theory)					
Course Code	:	CI248ATC	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
Natural Language Processing Category: Professional Core Elective (Only CS) (Theory)					
Course Code	:	CS248ATD	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
Linux Internals					
Category: Professional Core Elective (Common to CS, CI, CY and CD Programs) (Theory)					
Course Code	:	CS355TBF	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
Operation Research					
Category: Professional Core Elective					
(Only CS)					
(Theory)					
Course Code	:	CS365TDB	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	100 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	3 Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B		
(Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3	Unit - II: Compulsory	18



RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	CONTENTS	MARKS
4 & 5	Unit - III: Question 4 or 5	18
6 & 7	Unit - IV: Question 6 or 7	18
9 & 9	Unit - V: Question 8 or 9	18
TOTAL		100



Semester - IV					
National Service Scheme Category: AE Course (All programs) (Practical)					
Course Code	:	HS238LA	CIE	:	50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	1.5 Hours
Total credits:	:	1	SEE Duration	:	2 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	Presentation 1: Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data.	10
2	EXPERIENTIAL LEARNING: Presentation 2 (phase 2) Content development, strategies for implementation methodologies (10), Case Study-based Teaching-Learning (10), Sector wise study and consolidation (10), Video based seminar (4-5 minutes per student) (10), TOTAL MARKS ADDED UPTO 40 MARKS	40
MAXIMUM MARKS FOR THE CIE		50

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Implementation strategies of the project with report	50
TOTAL		50



Semester - IV					
National Cadet Corps Category: AE Course (All programs) (Practical)					
Course Code	:	HS238LB	CIE	:	50 Marks
Credit distribution L:T:P	:	0:0:1	SEE	:	50 Marks
Pedagogical Hours (L:T:P:S)	:	0:0:30:0	CIE Duration	:	1.5 Hours
Total credits:	:	1	SEE Duration	:	2 Hours
Hardware experiments Here you can put your instructions here. If you don't want this instruction, simply comment this line. 1. Design Innovative Experiments 2. [EL] For EL experiment					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	Presentation 1: Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data.	10
2	EXPERIENTIAL LEARNING: Presentation 2 (phase 2) Content development, strategies for implementation methodologies (10), Case Study-based Teaching-Learning (10), Sector wise study and consolidation (10), Video based seminar (4-5 minutes per student) (10), TOTAL MARKS ADDED UPTO 40 MARKS	40
MAXIMUM MARKS FOR THE CIE		50

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Implementation strategies of the project with report	50
TOTAL		50



Semester - IV					
Bridge Course C Programming Category: Bridge Course (Common to Common to CS,CD,CY and CI Programs) (Theory)					
Course Code	:	CS139AT	CIE	:	100 Marks
Credit distribution L:T:P	:	3:0:0	SEE	:	— Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45	CIE Duration	:	2 Hours
Total credits:	:	3	SEE Duration	:	— Hours
Unit - I					4 Hours
sample					
Course Outcomes: After completing the course, the students will be able to:					
1. sample					
References					
[1] B. Razavi, <i>Design of Analog CMOS Integrated Circuits</i> . McGraw-Hill Education, 2000, ISBN: 978-0072380323					

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE		100